INCH-POUND
MIL-M-70806(AR)
4 September 1990

SUPERSEDING
M60PD-M-62213(AT)
7 December 1978

MILITARY SPECIFICATION

MOTOR ASSEMBLY

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification supports the acquisition of the turret drive direct current electric motor assembly used in the M60A1 series tank. The motor assembly is a varying speed, 27.5 volt direct current (vdc), 20 horsepower (hp), series or compound wound, intermittent duty unit. This specification includes the minimum essential Engineering and Packaging Requirements and the necessary Quality Assurance Provisions to determine that these requirements have been met.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research, Development and Engineering Center, US Army Armament, Munitions and Chemical Command, ATTN: SMCAR-CCB-SAS, Watervliet, N.Y. 12189-4050 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6105

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SPECIFICATIONS

Military

MIL-H-46170 - Hydraulic Fluid, Rust-Inhibited, Fire-Resistant, Synthetic Hydrocarbon Base

MIL-P-14232 - Parts, Equipment and Tools for Army Material, Packaging and Packing of

MIL-S-45743 - Soldering, Manual Type, High Reliability Electrical and Electronic Equipment

STANDARDS

Military

DOD-STD-100 - Engineering Drawing Practices

MIL-STD-105 - Sampling Procedures for Tables for Inspection by Attributes

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of US Military Property

MIL-STD-193 - Painting Procedures and Marking for Vehicles

MIL-STD-202 - Test Methods for Electrical and Electronic Component Parts

MIL-STD-454 - Electronic Equipment, Standard General Requirements for

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurements of

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

MIL-STD-889 - Dissimilar Metals

MIL-STD-1275 - Electrical Circuit, 28 Volt DC Transient Characteristics for Military Vehicles

MIL-STD-45662 - Calibration System Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

Military

U.S. Army Armament Research, Development and Engineering Center

8724242 - Shell Assembly

10905300 - Accumulator

10959033 - Motor Endurance Test Cooling Device

11590942 - Switch, Pressure

11637230 - Power Pack Control Assembly

11655194 - Soldering, Electrical Connection for Electrical and Electronic Equipment, General Specification for

11668642 - Motor Assembly

11674112 - Power Pack Control Assembly

(Copies of drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Procuring Contracting Officer.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

National Electrical Manufacturers Association (NEMA)

NEMA MG 1 - Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturer's Association, 2101 L Street, NW, Washington, DC 20037).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article. The contractor shall submit a first article unless it is specifically waived in the contract (see 4.3 and 6.2). No first article requirements shall be waived without review and approval by the Procuring Contracting Officer.

3.2 Characteristics.

3.2.1 Performance. Unless otherwise specified, the assembly shall meet the performance requirements specified herein under the following conditions:

a. Temperature	-	73 ± 18 degrees Fahrenheit (°F)
b. Pressure	-	28.5 +2.0/-4.5 inches of mercury
c. Humidity	-	50 + 49/30 percent (%)
d. Voltage	-	27.5 ± 0.5 vdc
e. Cooling	-	Brushes cooled with an external device simulated by Motor Endurance Test Cooling Device 10959033

3.2.1.1 Nominal operation. The nominal operating characteristics of the motor assembly shall be as follows:

Voltage	-	27.5 volts DC
Current	-	340 amperes (amp)
Speed	-	3,400 revolutions per minute (rpm)
Horsepower Output	-	10 hp

3.2.1.2 Insulation resistance. The insulation resistance of the motor assembly shall not be less than the following:

a. Armature Circuit:	25 Megohms
b. Field Circuit:	50 Megohms

3.2.1.3 Low voltage. The motor assembly shall produce a minimum of 5.0 hp output, with 15.4 lb-ft minimum load, with an input voltage of 18 vdc.

3.2.1.4 Position. The motor assembly shall operate at full load speed without evidence of damage in any position.

3.2.1.5 Current. The motor assembly with an input voltage of 27.5 ± 0.5 vdc shall have a maximum allowable current draw at 6.5 pound-feet (lb-ft) and at 15.4 pound feet (lb-ft) torque as shown on FIGURES 1 and 2.

3.2.1.6 Efficiency. The motor assembly shall have an efficiency as shown on FIGURE 3.

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3.2.1.7 Speed range. The rated speed range shall be as shown on FIGURE 3, and be within the limits as shown in FIGURES 1 and 2 for 6.5 lb-ft and 15.4 lb-ft torque respectively.

3.2.1.8 Rotation. The rotation of the drive, as viewed from the power coupling end of the motor assembly, shall be clockwise with positive (normal) power supply connection polarity.

3.2.1.9 Horsepower and torque performance. The motor assembly horsepower and torque performance shall be as shown on FIGURE 3.

3.2.1.10 Overspeed. The motor assembly shall withstand a continuous speed of 6,000 rpm for 5 minutes without degradation.

3.2.1.11 Dielectric strength. There shall be no puncture of solid insulation, no continuing sparkover, no breakdown of dielectric material, and no damage to the insulation sufficient to cause failure of the motor assembly when the motor assembly is tested at 500 volts root mean square (vrms), 60 hertz (hz).

3.2.1.12 Dynamic balance. The motor assembly shall conform to the requirements of NEMA Standards Section MG1-23.52 and the maximum peak-to-peak displacement in any direction shall not exceed .002 inch.

3.2.1.13 Case temperature. The case temperature of the motor assembly shall not exceed 260°F upon completion of 5,000 cycles of operation (2,500 cycles at ambient temperature, then 2,500 cycles at 125 ± 5°F) with a duty cycle of 2.5 + 0.5/-0.0 seconds ON, 2.5 +0.5/-0.0 seconds OFF, an oil temperature of 205 ± 5°F and a load as specified in 3.2.2.3.3.

3.2.2 Physical characteristics.

3.2.2.1 Weight. The motor assembly shall have a maximum weight of 100 lbs.

3.2.2.2 Enclosure. The motor assembly shall be a watertight fan-cooled enclosure as described herein and depicted on Drawing 11668642.

3.2.2.3 Endurance.

3.2.2.3.1 Rapid cycling. With an input voltage of 25.5 ± 0.5 vdc, a load as specified in 3.2.2.3.3, an oil temperature of 205 ± 5°F, and an air flow over brushes as shown on Drawing 10959033 applied, the motor shall show no evidence of performance degradation per 3.2.1 after 5,000 cycles of operation (2,500 cycles at ambient temperature, then 2,500 cycles at 125 ± 5°F) with a duty cycle of: 2.5 +0.5/-0.0 seconds ON, 2.5 +0.5/-0.0 seconds OFF.

3.2.2.3.2 Normal cycling. With an input voltage of 25.5 ± 0.5 vdc, a load as specified in 3.2.2.3.3, an oil temperature of 180 ± 10°F, and an airflow over brushes as shown on Drawing 10959033 applied, the motor shall have no need for servicing or replacement of parts during 225,000 cycles of operation at a duty cycle of: 2.5 ± 0.5 seconds ON, 5.5 ± 0.5 seconds OFF. Post endurance performance change from pre-endurance performance per 3.2.1 shall not exceed 7.5%.

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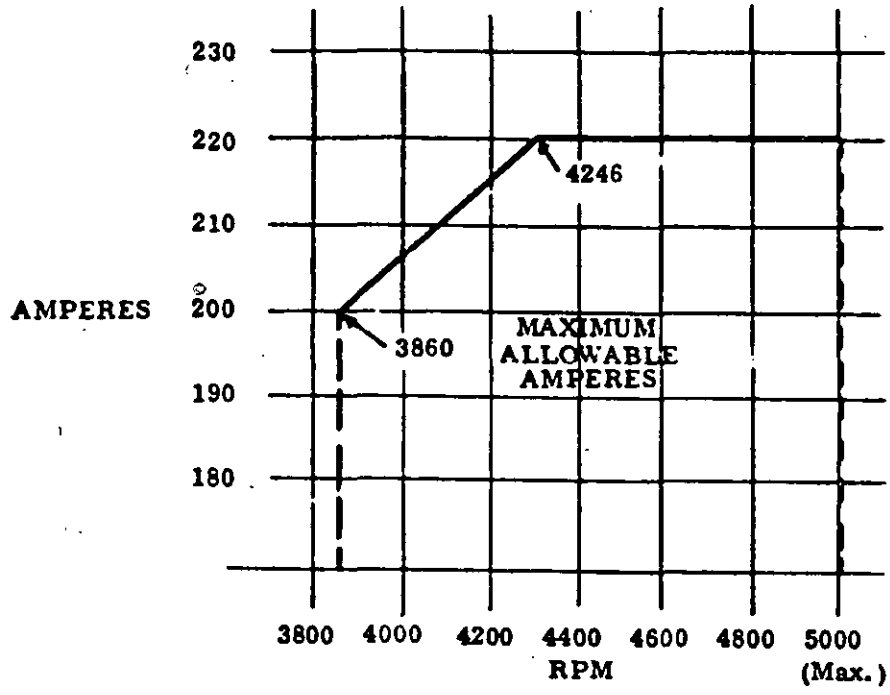


FIGURE 1. Load limit at 6.5 lb-ft

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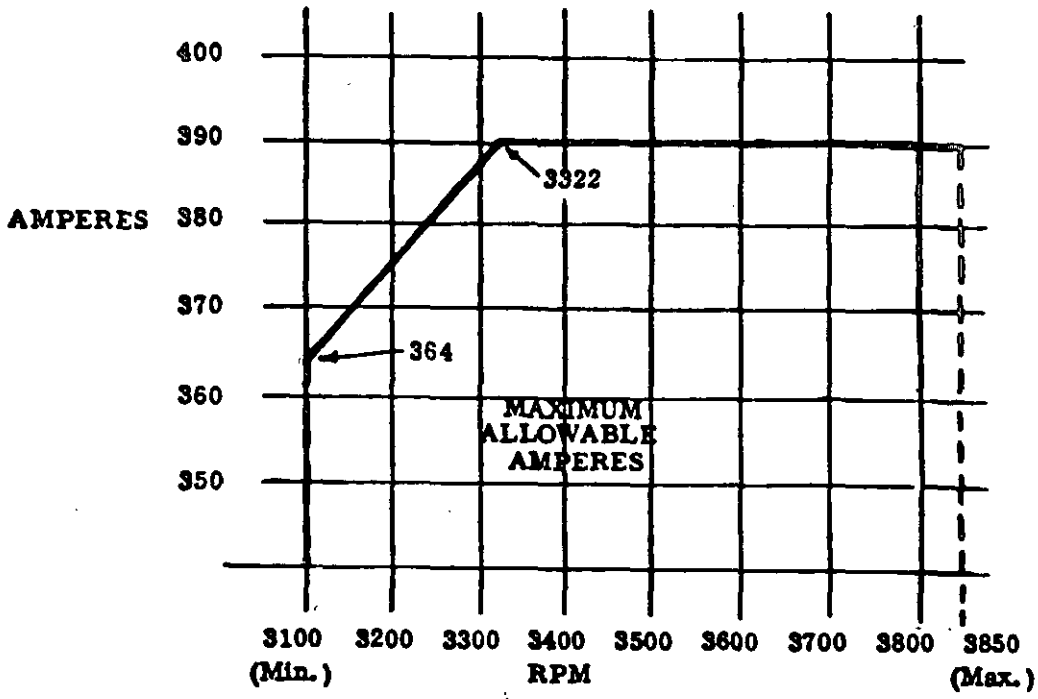


FIGURE 2. Load limit at 15.4 lb-ft

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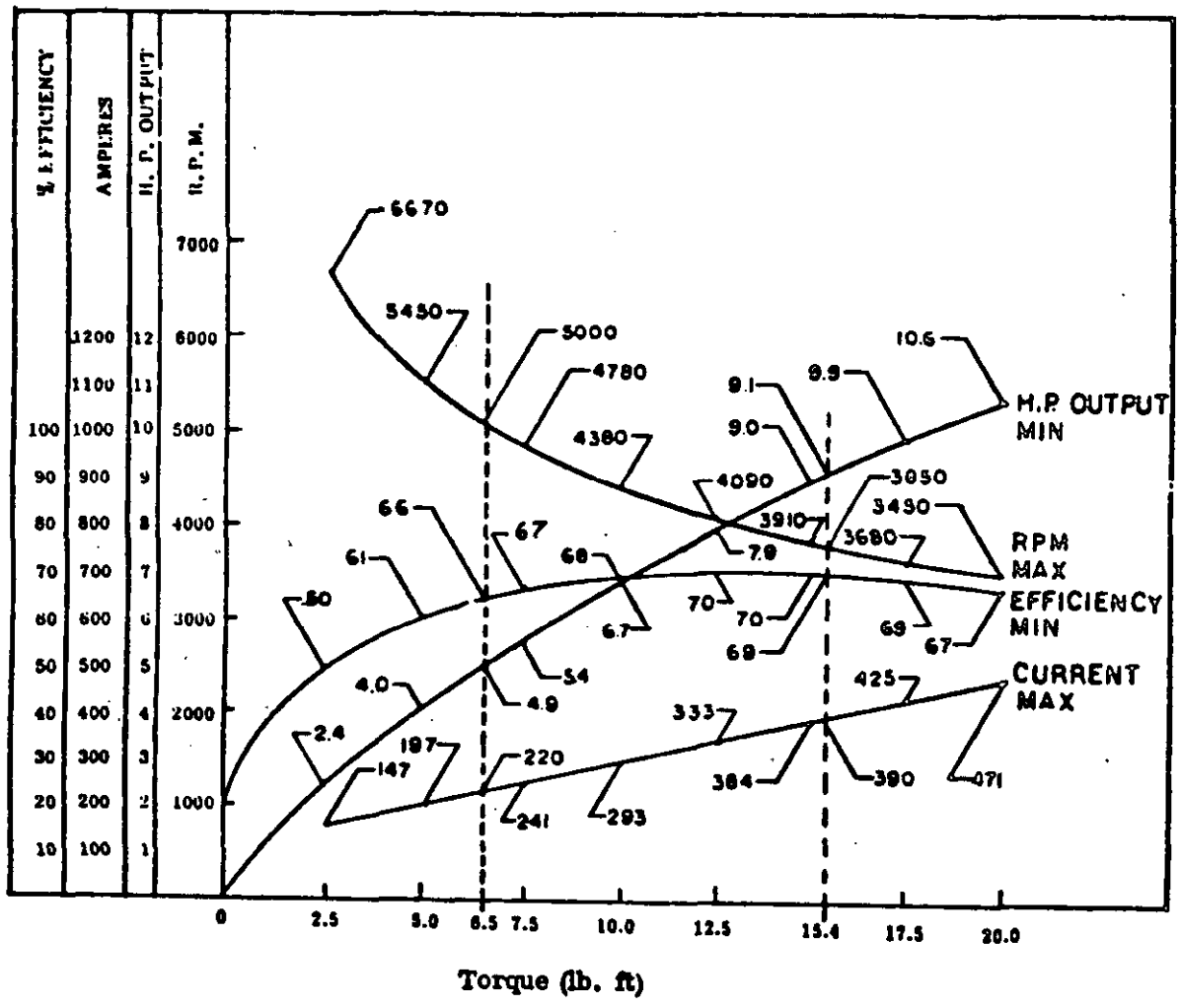


FIGURE 3. Characteristic curves

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3.2.2.3.3 Load. Motor assembly load for requirements as stated in 3.2.1.13, 3.2.2.3.1, and 3.2.2.3.2 is shown schematically in FIGURE 4 and represents a hydraulic pump with an inertia of $5.0 \pm 0.15 \times 10^{-5}$ ft-lb-sec^2.

3.2.3 Overhaul. The Mean Time to Overhaul (MTO) of the motor assembly shall not exceed 4 hours. Overhaul shall consist of disassembly, replacement of brushes, bearings, and seals, and functional test.

3.2.4 Environmental conditions.

3.2.4.1 Temperature. The motor assembly shall meet the performance requirements specified herein at temperatures ranging from -25°F to 125°F and shall not be damaged by storage conditions ranging from -65° to 160°F. Performance degradation at temperature extremes shall not exceed 7.5%.

3.2.4.2 Vibration. The motor assembly shall be capable of withstanding the vibration profile per FIGURE 6 for period of 80 minutes (including up to 4 resonance dwells of 13-1/3 minutes each) in each of the 3 mutually perpendicular axis, X, Y and Z, as shown in FIGURE 5.

3.2.4.3 Shock (nonoperating). The motor assembly shall be capable of withstanding exposure to three half sine wave shock pulses of 40 ± 4 gravity unit (g) peak for a duration of 18 ± 3 milliseconds applied in both directions along the three mutually perpendicular axes X, Y and Z, as shown in FIGURE 5.

3.2.4.4 Humidity. The motor assembly shall meet the requirements specified herein during and after exposure to humidity conditions up to 99% R.H.

3.2.4.5 Corrosion. The assembly shall be capable of operating as specified herein after exposure to a 5% sodium chloride atomized spray for 48 hours minimum.

3.2.4.6 Fungistatic materials. All non-metallic materials shall be inherently fungistatic or treated to resist fungus growth except when used for components of potted or otherwise moisture sealed assemblies.

3.2.4.7 Waterproofness. The motor assembly shall be capable of withstanding 6 psig while submerged in clear water at ambient temperature for two (2) minutes without evidence of leakage. The motor assembly shall be submerged with the electrical connector capped and the uppermost surface of the motor a minimum of one inch below the surface of the water. Motor shall not be electrically operated during immersion.

3.2.4.8 Transient voltage. The motor shall meet requirements during exposure to voltage as specified in MIL-STD-1275 (see 4.4.4.4.9).

3.3 Design and construction.

3.3.1 Materials, processes, and parts. Exposed functional parts shall be fabricated from suitable corrosion resistant material or treated to prevent corrosion. Exposed nonfunctional parts shall be painted white conforming to MIL-STD-193.

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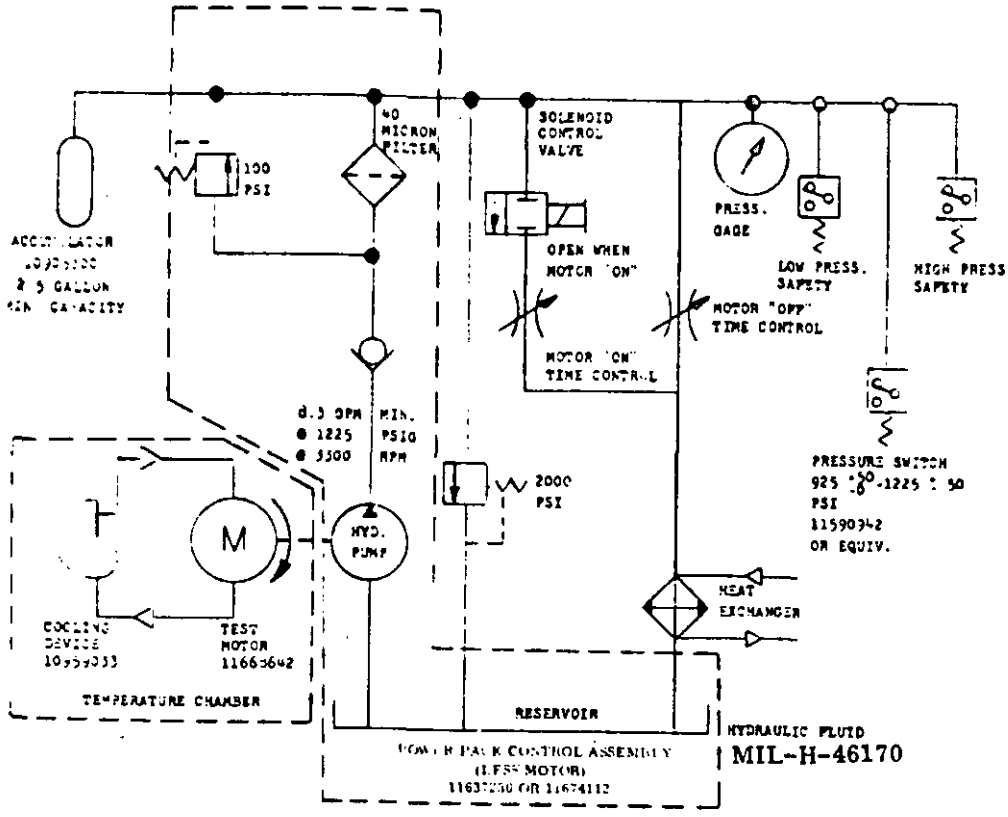


FIGURE 4A. Typical Electrical Motor Load Schematic

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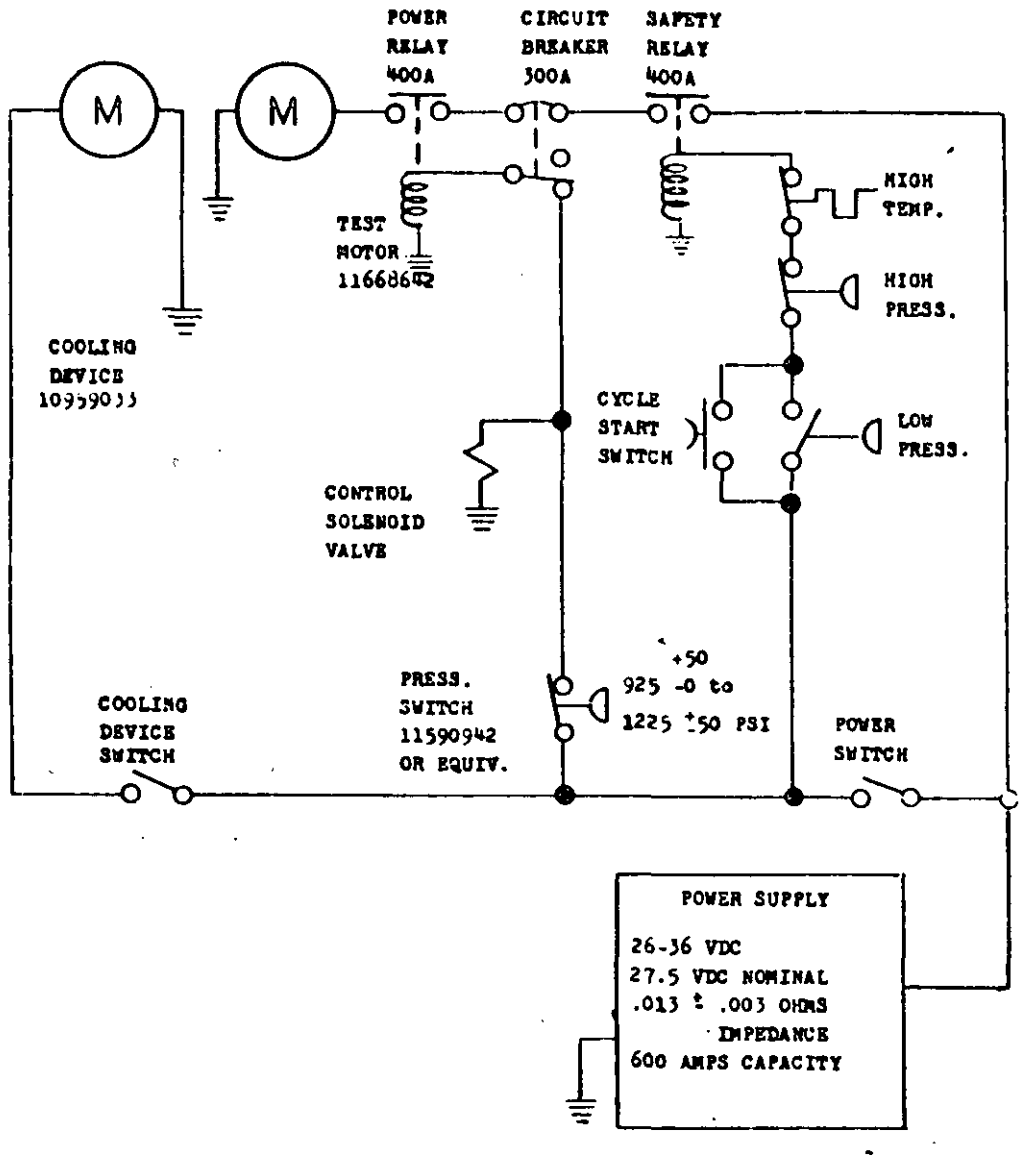


FIGURE 4B. Typical Electrical Motor Power Schematic

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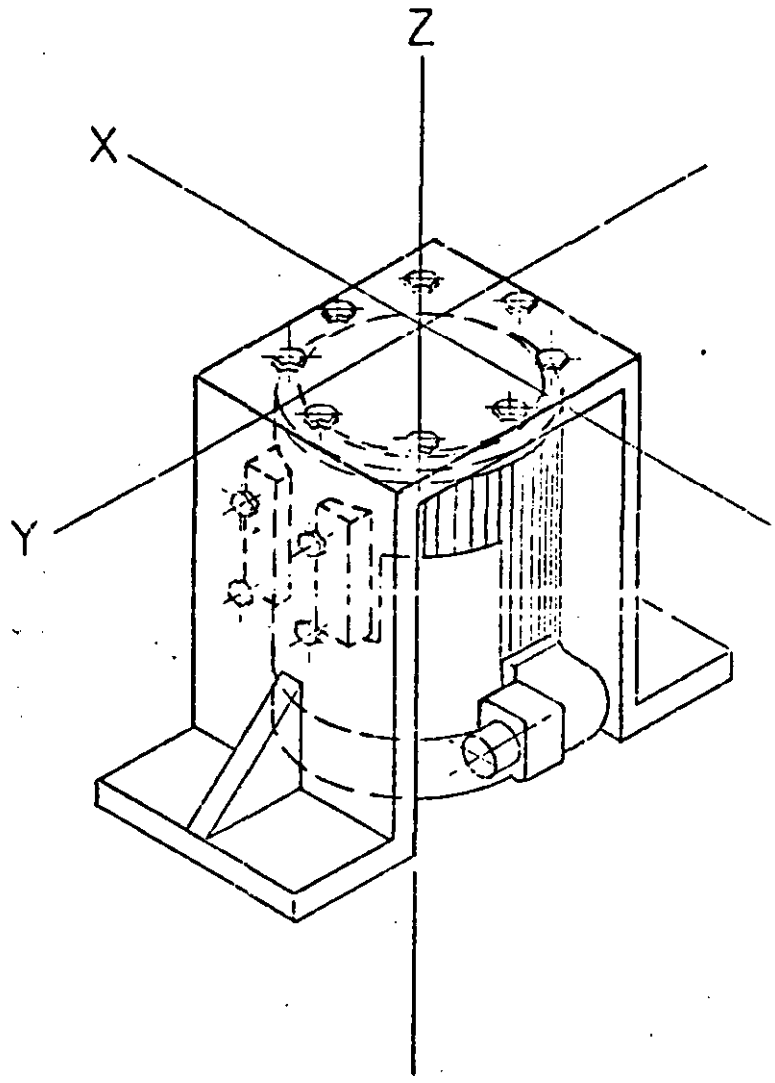


FIGURE 5. Shock and Vibration Axes

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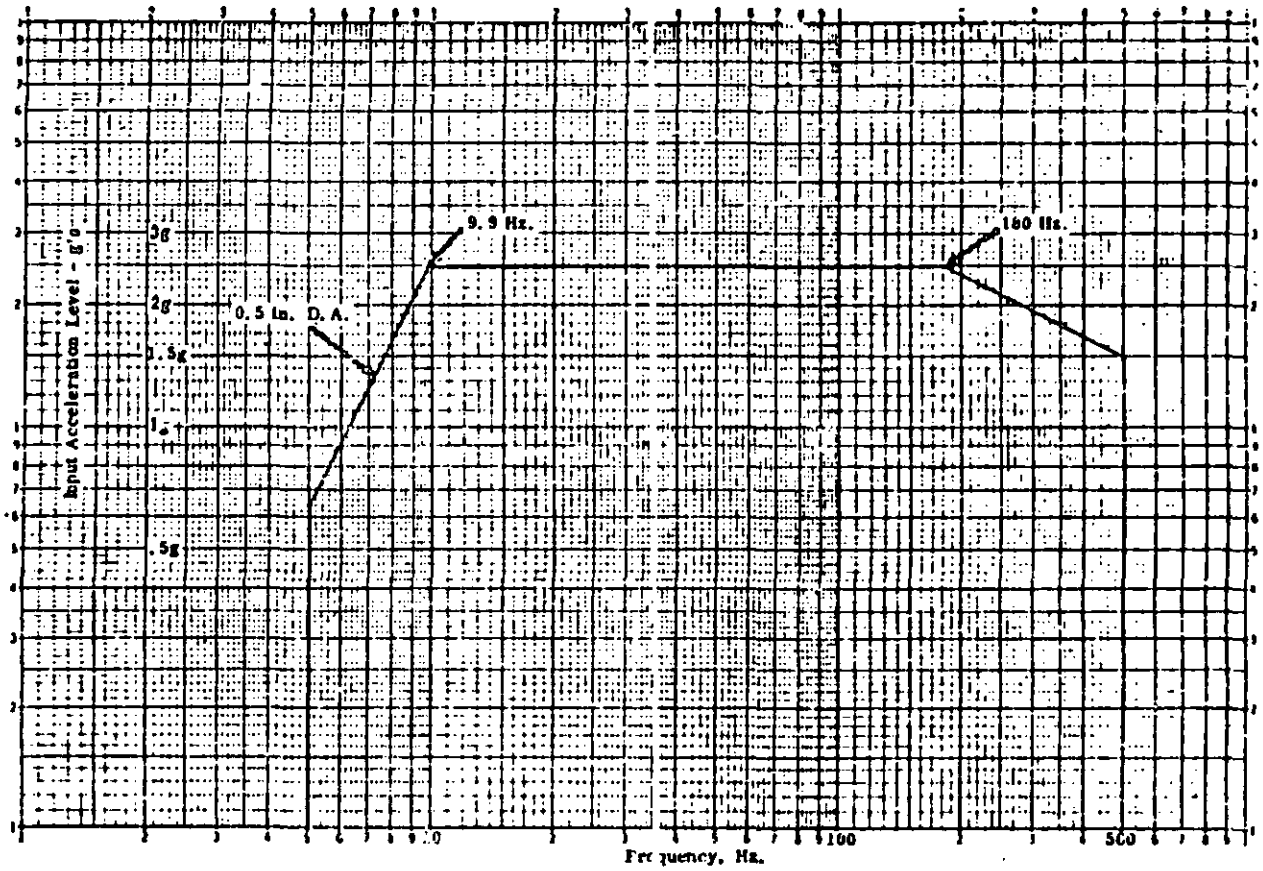


FIGURE 6. Vibration Profile

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3.3.1.1 Standards of design and manufacture. Unless otherwise specified and where applicable, the motor assembly shall be in accordance with the following requirements of MIL-STD-454.

Number	Subject
2	Capacitors
3	Flammable Materials
6	Bearings
11	Insulating Materials (Class II)
12	Fastener Hardware
15	Ferrous Alloys, Corrosion Resistance
19	Terminals
21	Castings
23	Adhesives
24	Welds, Resistance, Electrical Interconnections
26	Arc-Resistant Materials
59	Brazing

3.3.1.2 Mounting and support flanges. The mounting and support flanges shall conform to the motor assembly Drawing 11668642.

3.3.1.3 Shaft. A shaft extension with keyway shall conform to motor assembly Drawing 11668642.

3.3.1.4 Brush holders.

3.3.1.4.1 Accessibility. Brush holders shall be readily accessible for adjustment and renewal of brushes and springs.

3.3.1.5 Free end play. The free end play clearance of the shaft shall be .010 inches maximum.

3.3.1.6 Connections.

3.3.1.6.1 Securing connections. All connections liable to become loosened by vibration shall be provided with locking devices. Connections and leads shall be secured in a reliable manner to prevent their coming in contact with moving or stationary parts.

3.3.1.7 Connector, external. The connector must mate with and seal shell assembly 8724242.

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3.3.1.8 Class of insulation. The insulation shall be a minimum of Class H (180°C) per NEMA Standards.

3.3.1.9 Soldering. Electrical connections not welded or otherwise mechanically connected shall be soldered in accordance with MIL-S-45743.

3.3.1.10 Dissimilar metals. Except where necessary to complete an electrical circuit, contact between dissimilar metals, which would foster galvanic actions should be avoided. Where such contact is not necessary to complete an electrical circuit, but is otherwise unavoidable, parts shall be electrically insulated. Dissimilar metals are defined in MIL-STD-889.

3.3.2 Electro-magnetic interference. The motor assembly when operating per MIL-STD-462 under the intermittent duty cycle of 2.5 + 0.5 seconds ON and 2.5 ± 0.5 seconds OFF, shall meet the requirement of MIL-STD-461 for *electrical/electro-mechanical equipment.*

3.3.3 Identification and markings.

3.3.3.1 Permanent mark. The military part number and serial number shall be permanently inscribed on the frame in the location shown on Motor Assembly Drawing 11668642.

3.3.3.2 Nameplate. Nameplate markings shall conform to MIL-STD-130. The nameplate shall be firmly attached to the exterior of the motor in accordance with MIL-STD-130 and positioned as shown on Motor Assembly Drawing 11668642. The nameplate shall include the following minimum information:

a. Manufacturer's name, CAGE number, serial number, date of manufacture.

b. Nominal voltage, horsepower, type of winding, speed and duty.

c. Class of insulation.

d. Military part number.

e. Federal stock number (when assigned).

3.3.4 Workmanship. The motor assembly shall be free of any electrical, mechanical, or physical defect which would affect the reliability or performance of the power supply.

3.3.5 Interchangeability. All similar parts, including repair parts, of the motor assembly furnished on the same order or built to the same drawings, shall be strictly interchangeable without the necessity of further machining, selective assembly, or hand fitting of any kind.

3.3.6 Recycled, reclaimed, and virgin materials. Except when intended use of the item will be jeopardized by the use of reclaimed or recycled materials, in preparing new and revising or amending specifications, preparing activities will ensure that:

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a. There is no exclusion to the use of recovered materials.

b. There is no requirement that an item be manufactured from virgin materials.

c. Within 1 year from the date of issue by the Environmental Protection Agency of guidelines designating items which are or can be produced with recovered materials, specifications for such products require the use of recovered materials to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.1.1 Inspection equipment. Inspection equipment or measuring setups must be capable of repetitive measurements to an accuracy of 10 percent of the component tolerance. Calibration of inspection and test equipment shall be in accordance with MIL-STD-45662. Where compliance with MIL-STD-45662 is not possible, special calibration requirements shall be prepared for approval by the Procuring Contracting Officer.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. Unless otherwise specified in the contract (see 6.2), first article motor assemblies shall be inspected by the contractor and submitted to the Government for approval. First article inspection shall be performed in accordance with TABLES I and II. First article motor assemblies shall be representative of the production processes to be used during quantity production.

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TABLE I. SEQUENCE AND INSPECTIONS APPLIED TO EACH SAMPLE FOR FIRST ARTICLE INSPECTION

Test Sequence	Paragraph Number	First Article Sample		
		1	2	3
Design and construction (except 4.4.4.5.2)	4.4.4.5	x	x	x
Physical characteristics (except 4.4.4.2.3)	4.4.4.2	x	x	x
Performance (except 4.4.4.1.13)	4.4.4.1	x	x	x
High temperature	4.4.4.4.1	x	x	x
Low temperature	4.4.4.4.2	x	x	x
Waterproofness	4.4.4.4.8	x	x	x
Endurance/Rapid cycling	4.4.4.2.3.1		x	x
Case temperature	4.4.4.1.13		x	x
Endurance/Normal cycling	4.4.4.2.3.2		x	x
Shock	4.4.4.4.4	x	x	
Vibration	4.4.4.4.3	x	x	
Humidity	4.4.4.4.5	x		
Corrosion	4.4.4.4.6	x		
Electromagnetic interference	4.4.4.5.2		x	
Packaging	4.4.3	x		

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TABLE II. CATEGORY OF INSPECTION

Description	Requirement Paragraph	Inspection Paragraph
Performance	3.2.1	4.4.4.1
Nominal operation	3.2.1.1	4.4.4.1.1
Insulation resistance	3.2.1.2	4.4.4.1.2
Low voltage	3.2.1.3	4.4.4.1.3
Position	3.2.1.4	4.4.4.1.4
Current	3.2.1.5	4.4.4.1.5
Efficiency	3.2.1.6	4.4.4.1.6
Qualification and control		4.4.4.1.6.1
Acceptance		4.4.4.1.6.2
Speed range	3.2.1.7	4.4.4.1.7
Qualification inspection		4.4.4.2.7.1
Acceptance		4.4.4.2.7.2
Rotation	3.2.1.8	4.4.4.1.8
Horsepower and torque perf.	3.2.1.9	4.4.4.1.9
Qualification and control		4.4.4.1.9.1
Acceptance		4.4.4.1.9.2
Overspeed	3.2.1.10	4.4.4.1.10
Dielectric strength	3.2.1.11	4.4.4.1.11
Dynamic balance	3.2.1.12	4.4.4.1.12
Case temperature	3.2.1.13	4.4.4.1.13
Physical characteristics	3.2.2	4.4.4.2
Weight	3.2.2.1	4.4.4.2.1
Enclosure	3.2.2.2	4.4.4.2.2
Endurance	3.2.2.3	4.4.4.2.3
Rapid cycling	3.2.2.3.1	4.4.4.2.3.1
Normal cycling	3.2.2.3.2	4.4.4.2.3.2
Qual. pre. & initial inspection		4.4.4.2.3.2.1
Control		4.4.4.2.3.2.2
Load	3.2.2.3.3	4.4.4.2.3.3
Overhaul	3.2.3	4.4.4.3

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TABLE II. CATEGORY OF INSPECTION

Description	Requirement Paragraph	Inspection Paragraph
Environmental Conditions	3.2.4	4.4.4.4
High temperature	3.2.4.1	4.4.4.4.1
Low temperature	3.2.4.1	4.4.4.4.2
Vibration	3.2.4.2	4.4.4.4.3
Shock	3.2.4.3	4.4.4.4.4
Humidity	3.2.4.4	4.4.4.4.5
Corrosion	3.2.4.5	4.4.4.4.6
Fungistatic materials	3.2.4.6	4.4.4.4.7
Waterproofness	3.2.4.7	4.4.4.4.8
Design and Construction	3.3	4.4.4.5
Materials, processes, and parts	3.3.1 thru 3.3.1.10	4.4.4.5.1
Electro-magnetic Interference	3.3.2	4.4.4.5.2
Identification marking	3.3.3 thru 3.3.3.2	4.4.4.5.3
Workmanship	3.3.4	4.4.4.5.4
Interchangeability	3.3.5	4.4.4.5.5
Packaging	5.	4.4.3

NOTES: All requirements must be inspected for first article inspection (see TABLE I.). Quality conformance inspection shall be according to the sampling plan specified in the contract. For suggested sampling parameters, see 6.3.

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4.3.1 First article failure. Failure of a first article assembly during or as a result of first article inspections shall be cause for rejection of the assembly. The Procuring Contracting Officer shall refuse acceptance of production assemblies until evidence of corrective action is provided.

4.4 Quality conformance inspections.

4.4.1 Lot by lot inspection. Lot by lot inspection shall consist of sampling or 100% acceptance inspection as specified in the contract or purchase order. TABLE III (see 6.3) contains suggested sampling parameters. An inspection lot shall consist of all assemblies of one type, submitted at one time for Quality Conformance Inspection.

4.4.1.1 Sampling. Sampling and inspection shall be conducted in accordance with MIL-STD-105 on the basis of percent defective for those characteristics of TABLE II assigned an Acceptance Quality Level (AQL). Characteristics having the same AQL shall be treated as a group.

4.4.1.1.1 AQL validation. Before sampling can commence for any production contract, a minimum of twenty assemblies shall be subjected to 100% inspection to verify conformance to requirements listed in TABLE II. Process average for each requirement shall be computed as specified below. If the computed process average for the requirements exceeds the specified AQL, 100% inspection shall be continued until the process average for twenty consecutive assemblies is less than the specified AQL.

$$\text{Process average} = \frac{\text{Number of Defectives}}{\text{Number of Assemblies Inspected}} \times 100$$

4.4.1.1.2 Sampling failures. Rejected assemblies or lots shall be processed in accordance with the acceptance and rejection criteria of MIL-STD-105.

4.4.1.2 Acceptance (100%) inspection. For the requirements specified for acceptance (100%) inspection in the contract or purchase order, each assembly of the inspection lot shall be subjected to the tests specified therein. Inspection shall be performed by the supplier at the place of manufacture except as specified in 4.1.

4.4.1.2.1 Acceptance (100%) inspection failures. Any assembly that fails to conform to any acceptance (100%) inspection shall be rejected. The rejected assembly may be repaired or corrected and resubmitted for inspection.

4.4.2 Control inspection (see 6.2 and 6.3). Unless otherwise specified, control inspection shall be conducted by the supplier for maintaining control of the requirements whose length, expense, or destructivensss prohibit their inclusion under lot by lot inspection. Control inspection assemblies shall be selected at the frequencies specified in contract or purchase order and subjected to inspections to verify conformance to the requirements in TABLE II. These inspections need not be performed on the first control inspection quantity when first article inspection has been satisfactorily accomplished.

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4.4.3 Packaging inspection. The assembly shall be inspected by the contractor prior to shipment to ensure conformance to the preservation, packaging, packing and marking requirements in Section 5. Sampling inspections shall be limited to those characteristics that can be determined by visual examination.

4.4.4 Test methods and procedures.

a. Requirements for the following tests are as listed in TABLE II.

b. Unless otherwise specified, all examinations and tests shall be made under the following conditions:

(1) Temperature - 73 ± 18 degrees Fahrenheit (°F)

(2) Pressure - 28.5 +2.0/-4.5 inches of mercury

(3) Humidity - 50 ± 30 percent (%)

(4) Voltage - 27.5 ± 0.5 vdc

(5) Cooling - Brushes cooled with an external device simulated by Motor Endurance Test Cooling Device 10959033

c. Prior to the acceptance tests of 4.4.4.1, perform the waterproofness test of 4.4.4.4.8.

d. Where testing while operating under a load is required, the motor assembly shall be coupled to a suitable loading device, capable of loading it over its rated load and speed range. Provisions shall be made for monitoring torque, rotational speeds, input current, input voltage at the motor terminals, and temperature parameters.

e. Perform the insulation resistance tests of 4.4.4.1.2 and the dielectric strength test of 4.4.4.1.11 before the final assembly. For acceptance, objective evidence shall be provided attesting that assembly has met the insulation resistance and dielectric strength requirements.

4.4.4.1 Performance tests.

4.4.4.1.1 Nominal operation. Unless otherwise specified, input voltage for performance tests shall be 27.5 ± 0.5 vdc. A blower assembly (Drawing 10959033) shall be connected during applicable performance tests.

4.4.4.1.2 Insulation resistance. Prior to final assembly, stablilize the motor assembly at 77°F ± 10°F and measure the insulation resistance between

(1) the positive lead and frame ground for each field circuit, and

(2) between commutator and shaft for the armature circuit in accordance with MIL-STD-202, Method 302, Test Condition A. Verify that the resistance is not less than 50 megohms for each field circuit and not less than 25 megohms for the armature circuit which includes insulated brush holders.

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4.4.4.1.3 Low voltage. Measure the speed and torque output of the motor assembly while operating with a power source of 18 vdc for a period of 4.0 + 1.0 minute. Verify the torque output is 15.4 lb-ft minimum and the horsepower is 5.0 H.P. minimum.

4.4.4.1.4 Position. In conjunction with the test of 4.4.4.4.3, the motor assembly shall be operated with the output shaft up, down, and horizontal. Verify that there is no evidence of damage.

4.4.4.1.5 Current. Subject the motor assembly to 27.5 + 0.5 vdc at 6.5 lb-ft torque load. repeat with a 15.4 lb-ft torque load. Verify that the maximum allowable current conforms to FIGURES 1 and 2.

4.4.4.1.6 Efficiency.

4.4.4.1.6.1 Qualification and control. Subject the motor assembly to a load torque varied from 2.5 to 20.0 lb-ft in increments of 2.5 lb-ft. Measure and record the motor current, applied voltage, and rotational speed. Calculate the motor assembly's efficiency using the following formula:

$$\% \text{ EFF} = 14.2 \frac{\text{Torque (lb-ft) X RPM}}{\text{Volts X Amps}}$$

Verify that calculated efficiencies at each applied torque load equals or exceeds the efficiency curve in FIGURE 3.

4.4.4.1.6.2 Acceptance. Subject the motor assembly to torque loads of 6.5 lb-ft and 15.4 lb-ft and proceed as in 4.4.4.1.6.1. Verify that the calculated efficiency equals or exceeds that indicated in FIGURE 3.

4.4.4.1.7 Speed range.

4.4.4.1.7.1 Qualification inspection. Perform the speed range test in conjunction with 4.4.4.1.6.1. Verify that the speeds, in revolutions per minute at the different torque levels are equal to or less than the speed curve in FIGURE 3.

4.4.4.1.7.2 Acceptance. Verify that the speed at 6.5 lb-ft and 15.4 lb-ft torque loads of test 4.4.4.1.6.2 are equal to or less than the speed curve in FIGURE 3.

4.4.4.1.8 Rotation. Verify that the shaft directional rotation, as viewed from the power coupling end of the motor assembly, is clockwise with positive (normal) power supply connection polarity.

4.4.4.1.9 Horsepower and torque performance.

4.4.4.1.9.1 Qualification and control. Using the data obtained in 4.4.4.1.6.1, calculate the motor assembly's horsepower performance using the following formula:

$$\text{HP} = \frac{\text{Torque (lb-ft) X RPM}}{5{,}250}$$

Verify that calculated horsepower at each applied torque load will equal or exceed the horsepower curve in FIGURE 3.

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4.4.4.1.9.2 Acceptance. Calculate the motor assembly's horsepower performance for 6.5 lb-ft and 15.4 lb-ft torque data obtained in 4.4.4.1.6.2. The horsepower shall equal or exceed that indicated in FIGURE 3.

4.4.4.1.10 Overspeed. Subject the motor assembly to 6,000 rpm for a period of 5.0 + 1.0 minute. Verify that the motor assembly is operating smoothly, balanced, and without distortion or noise.

4.4.4.1.11 Dielectric strength. Prior to final assembly perform the dielectric strength test on the armature and field windings separately in accordance with MIL-STD-202, Method 301. Apply 500 vrms at a frequency of 60 Hz for a period of 75 + 5 seconds. Verify that there is no puncture of solid insulation, no continuing sparkover, and no breakdown of dielectric material.

4.4.4.1.12 Dynamic balance. Place the motor assembly on an elastic mounting in an operating position. The weight of the motor assembly should compress an elastic mounting approximately 0.25 inch, or position the motor assembly such that the natural frequency is one quarter or less of the operating rpm. Operate the motor assembly at reduced voltage, full load speed. Measure the amplitude of running vibration at the drive end in three axes and verify that it does not exceed 0.002 inch displacement.

4.4.4.1.13 Case temperature. This test shall be conducted in conjunction with the test in 4.4.4.2.3.1. During the last 5 minutes of the test, measure the temperature on the hottest spot of the motor case (excluding that part of the case that is enclosed within the pump/reservoir cavity) and verify that it does not exceed 260°F.

4.4.4.2 Physical characteristics.

4.4.4.2.1 Weight. Weight the motor assembly to verify that it does not exceed 100 pounds.

4.4.4.2.2 Enclosure. Visually examine the motor assembly for conformance to Drawing 11668642. Verify that the motor assembly is enclosed as specified.

4.4.4.2.3 Endurance.

4.4.4.2.3.1 Rapid cycling. Subject the motor assembly to a voltage 25.5 + 0.5 vdc, a load as specified in FIGURE 4 with an oil temperature of 205 + 5°F and an airflow over brushes as shown on Drawing 10959033. With the oil temperature stabilized, operate the motor assembly at a duty cycle of 2.5 +0.5/-0.0 seconds ON and 2.5 +0.5/-0.0 OFF. Upon reaching 2,000 cycles and without cyclic interruption, increase the motor assembly ambient temperature to 125 + 5°F within 45 minutes and continue to cycle the motor assembly until 5,000 cycles have been reached. Return the motor assembly to room temperature and subject the assembly to the performance tests of 4.4.4.1.3 through 4.4.4.1.10 and 4.4.4.1.12 and 4.4.4.1.13.

4.4.4.2.3.2 Normal cycling.

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4.4.4.2.3.2.1 Qualification, preproduction and initial production. Subject the motor assembly to voltage of 25.5 ± 0.5 vdc, a load as specified in FIGURE 4 with an oil temperature of 180°F ± 10°F, and an airflow over the brushes as shown on Drawing 10959033. Verify that the motor assembly operates for 225,000 cycles at duty cycle of: 2.5 ± 0.5 second On and 5.5 ± 0.5 second OFF. Post endurance change from pre-endurance performance tests of 4.4.4.1.3 through 4.4.4.1.10 and 4.4.4.1.12 shall not exceed 7.5%.

4.4.4.2.3.2.2 Control. This test shall be the same as 4.4.4.2.3.2.1, except that the motor shall operate for 168,500 cycles.

4.4.4.2.3.3 Load. Endurance tests shall be performed with the motor assembly operating in a test set-up as shown in FIGURE 4. The hydraulic pump of FIGURE 4 shall represent an inertia of $5.00 \pm .15 \times 10^{-5}$ ft-lb-sec^2.

4.4.4.3 Overhaul. To determine conformance to 3.2.3, verify that the motor assembly has a Mean-Time-To-Overhaul not greater than 4 hours. Overhaul shall consist of disassembly, replacement of brushes, bearings, seals, and functional test.

4.4.4.4 Environmental conditions.

a. Units undergoing qualification inspections of TABLE II shall be subjected to performance tests before, during (where required in the test method) and immediately after each environmental test performed.

b. Units undergoing conformance control inspections of TABLE II shall have passed the acceptance tests as indicated therein. After all environmental tests have been completed, the units shall be tested only once for the post environmental performance tests specified.

c. Performance tests required during environmental exposure and subsequent to environmental exposure are as follows:

(1) Low voltage - 4.4.4.1.3

(2) Current - 4.4.4.1.5

(3) Efficiency - 4.4.4.1.6.2

(4) Horsepower performance - 4.4.4.1.9.2

4.4.4.4.1 High temperature. Subject the motor assembly to the high temperature test specified in MIL-STD-810, Method 501, Procedure II, except storage temperature of step 4 shall be 160°F. At the conclusion of the storage temperature test stabilize the motor assembly at 125°F and test it in accordance with 4.4.4.4.c. Performance degradation at 125°F shall not exceed 7.5%. Return the motor assembly to room ambient temperature and test it in accordance with 4.4.4.4.c.

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4.4.4.4.2 Low temperature. Subject the motor assembly to the low temperature test specified in MIL-STD-810, Method 502, Procedure 1. Maintain the storage temperature at -65°F for a period of 12 hours minimum. At the conclusion of this time, stabilize the motor assembly at -25°F and test it in accordance with 4.4.4.4.c. Performance degradation at -25°F shall not exceed 7.5%. Return the motor assembly to room ambient temperature and test it in accordance with 4.4.4.4.c.

4.4.4.4.3 Vibration. The assembly shall be subjected to the vibration test specified in MIL-STD-810, Method 514, Procedure VIII, except as follows:

a. Test level shall be the vibration curve shown in figure 6. Time schedule shall be 40 minutes per axis (figure 5) including 6 2/3 minutes dwell time at each resonance.

b. Connections and instrumentation shall be attached to the assembly to permit operation and testing as specified in c through g.

c. Conduct resonant frequency search along each axis at room temperature and with input power applied over the frequency range of 5 to 500 Hz. Up to four resonant frequencies shall be identified (the four most severe resonant frequencies if more than four are present) in each axis for the resonant dwell test.

d. Place the assembly in a 125°F ambient temperature chamber and apply sufficient input power to operate the motor at 3,400 ± 300 RPM at no load.

e. Conduct sinusoidal vibration cycling and resonant dwell tests at 125°F and at the test levels and time schedules identified in "a" above.

f. Repeat the test described in "d" and "e" above except the ambient temperature shall be -25°F.

g. At the conclusion of the vibration test of "f" above return the motor assembly to room temperature and subject it to the performance tests of 4.4.4.4.c.

4.4.4.4.4 Shock. Mount the motor assembly as specified in FIGURE 5 and apply three half sine wave pulses in both directions in each of 3 mutually perpendicular axis, X, Y and Z, in accordance with MIL-STD-810, Method 516, Procedure I for a total of 18 shock pulses. Peak amplitude shall be 40g ± 4g with a time duration of 18 ± 3 msec measured at the 10 percent amplitude points. At the conclusion of this test, test the motor assembly in accordance with 4.4.4.4.c.

4.4.4.4.5 Humidity. Place the motor assembly in a humidity chamber and test it in accordance with MIL-STD-810, Method 507, Procedure II. During exposure, apply sufficient input power to operate the motor at 3400 ± 300 rpm at no load. At the conclusion of this test, test the motor assembly in accordance with 4.4.4.4.c.

4.4.4.4.6 Corrosion. Subject the motor assembly to the corrosion resistance test specified in MIL-STD-810, Method 509. At the conclusion of this test, test the motor assembly in accordance with 4.4.4.4.c.

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4.4.4.4.7 Fungistatic materials. Certification of compliance shall be provided attesting that the motor assembly is constructed of materials that will not support fungus growth.

4.4.4.4.8 Waterproofness. The motor assembly shall be submerged in a container with the uppermost surface a minimum of one inch below the surface of the water. Subject the unit to a pressure of 6 psig (internal pressure). (A modified brush cover plate may be used for pressurization location and the electrical connector shall be capped). The pressure shall be maintained for a two minute period without evidence of leakage. Bubbles which are the result of entrapped air on exterior surfaces of the assembly shall not be considered a leak. After exposure, the unit shall pass the performance tests of 4.4.4.1.

4.4.4.4.9 Transient voltage. To determine conformance to 3.2.4.8, the motor shall be subjected to transient voltage as specified in MIL-STD-1275.

4.4.4.5 Design and construction.

4.4.4.5.1 Materials, processes, and parts. To determine conformance to 3.3.1, unless otherwise specified, examine the motor assembly for conformance to MIL-STD-454 requirements as specified. Mounting, support flanges, shaft extension, and keyway shall be examined for conformance to Drawing 11668642. Measure the shaft end play clearance and verify that it is 0.010 inches maximum. Visually examine the external connector for mating with and sealing shell assembly 8724242. Examine connectors and leads for clearance and locking devices. Certification shall be provided attesting that soldering has been accomplished per specification 11655194. Dissimilar metals shall not be used in intimate contact, unless suitably protected against electrolytic corrosion as defined in MIL-STD-889.

4.4.4.5.2 Electromagnetic interference. To determine conformance to 3.3.2, subject the motor assembly to the electromagnetic interference tests specified in MIL-STD-462, Methods CE04 and RE02. Use a duty cycle of 2.5 + 0.5 second ON and 2.5 + 0.5 second OFF . At the conclusion of this test a certification shall be provided attesting that the motor assembly meets the electrical/electro-mechanical requirements of MIL-STD-461.

4.4.4.5.3 Identification and markings. To determine conformance to 3.3.3, visually examine the motor assembly for conformance to MIL-STD-130. Verify that the permanent marks and nameplates are as specified.

4.4.4.5.4 Workmanship. To determine conformance to 3.3.4, examine each item of the inspection lot at all phases of inspection testing and verify conformance with MIL-STD-454, Requirement 9.

4.4.4.5.5 Interchangeability. To determine conformance to 3.3.5, a certification of compliance shall be provided attesting that the motor assembly 11668642 is in conformance with interchangeability requirements of DOD-STD-100.

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5. PACKAGING

5.1 General. Levels of preservation, packaging and packing of the assembly shall be specified by the procuring activity (see 6.2). Definition of preservation, packaging and packing levels (A, B or C) are specified in 5.1.1 and 5.1.2.

5.1.1 Preservation and packaging.

LEVEL A. Preservation and packaging which will afford adequate protection against corrosion, deterioration and damage during shipment, handling, indeterminate storage and world-wide redistribution.

LEVEL B. Preservation and packaging which will afford adequate protection against corrosion, deterioration and damage during reshipment, handling and known storage conditions for periods normally not exceeding a year.

LEVEL C. Preservation and packaging which will afford adequate protection against corrosion, deterioration and damage during shipment from supply source to the first receiving activity for immediate use (vendor to user).

5.1.2 Packing.

LEVEL A. Packing which will afford adequate protection during shipment, handling, indeterminate storage, and world-wide redistribution.

LEVEL B. Packing which will afford adequate protection against domestic shipment, handling, and covered storage.

LEVEL C. Packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.2 Preservation, packaging, and packing materials. Unless otherwise specified, the materials used in preservation, packaging and packing of the assembly shall be as specified in MIL-P-14232.

5.3 Preservation and packaging. Unless otherwise specified by the contract or purchase order (see 6.2), preservation and packaging of the assembly shall be in accordance with level A, B, or C of specification MIL-P-14232. The internal cleanliness of the assembly, prior to being capped for preservation and packaging, shall meet the cleanliness levels specified in 3.2.2.2.

5.4 Packing. Unless otherwise specified by the contract or purchase order (see 6.2), packing of the assembly will be in accordance with level A, B, or C of specification MIL-P-14232.

5.5 Marking of shipments. Marking shall be specified in the contract or purchase order (see 6.2). In the absence of specific instructions, interior packages and exterior shipping containers shall be marked in accordance with standard MIL-STD-129.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The motor assembly is intended for use on the M60A1 series tank to rotate the turret and elevate or depress the main gun.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

c. First article tests required (see 4.3), the sampling parameters required (see 6.3), and the activity responsible for testing and its facility.

NOTE: Motor assemblies subjected to these tests, which are destructive in nature, shall not be delivered for use as defined in 6.1 and shall be marked: DO NOT USE.

d. Applicable levels of packaging and packing (see 5.3 and 5.4) and marking instructions (see 5.5).

e. Additional motor assemblies required to perform control test (see 4.4.2).

NOTE: Units subjected to destructive control tests (4.4.4.2.3, 4.4.4.4.3 and 4.4.4.4.4) shall not be delivered for use as defined in 6.1, and shall be indelibly marked: DO NOT USE.

6.3 Sampling parameters for quality conformance inspection shall be as specified in the contract. Suggested sampling parameters are indicated in TABLE III.

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TABLE III. SUGGESTED SAMPLING PARAMETERS.

Description	Requirement Paragraph	Inspection Paragraph	Notes
Performance	3.2.1	4.4.4.1	
Nominal operation	3.2.1.1	4.4.4.1.1	
Insulation resistance	3.2.1.2	4.4.4.1.2	x
Low voltage	3.2.1.3	4.4.4.1.3	x
Position	3.2.1.4	4.4.4.1.4	
Current	3.2.1.5	4.4.4.1.5	x
Efficiency	3.2.1.6	4.4.4.1.6	
Qualification and control		4.4.4.1.6.1	b(1/50)
Acceptance		4.4.4.1.6.2	x
Speed range	3.2.1.7	4.4.4.1.7	
Qualification inspection		4.4.4.2.7.1	
Acceptance		4.4.4.2.7.2	x
Rotation	3.2.1.8	4.4.4.1.8	x
Horsepower and torque perf.	3.2.1.9	4.4.4.1.9	
Qualification and control		4.4.4.1.9.1	b(1/50)
Acceptance		4.4.4.1.9.2	x
Overspeed	3.2.1.10	4.4.4.1.10	b(1/50)
Dielectric strength	3.2.1.11	4.4.4.1.11	
Dynamic balance	3.2.1.12	4.4.4.1.12	x
Case temperature	3.2.1.13	4.4.4.1.13	
Physical characteristics	3.2.2	4.4.4.2	
Weight	3.2.2.1	4.4.4.2.1	
Enclosure	3.2.2.2	4.4.4.2.2	a(2.5)
Endurance	3.2.2.3	4.4.4.2.3	
Rapid cycling	3.2.2.3.1	4.4.4.2.3.1	b(1/200)
Normal cycling	3.2.2.3.2	4.4.4.2.3.2	
Qual. pre. & initial inspection		4.4.4.2.3.2.1	
Control		4.4.4.2.3.2.2	b(1/200)
Load	3.2.2.3.3	4.4.4.2.3.3	
Overhaul	3.2.3	4.4.4.3	

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TABLE III. SUGGESTED SAMPLING PARAMETERS.

Description	Requirement Paragraph	Inspection Paragraph	Notes
Environmental Conditions	3.2.4	4.4.4.4	
High temperature	3.2.4.1	4.4.4.4.1	
Low temperature	3.2.4.1	4.4.4.4.2	
Vibration	3.2.4.2	4.4.4.4.3	b(1/200)
Shock	3.2.4.3	4.4.4.4.4	b(1/200)
Humidity	3.2.4.4	4.4.4.4.5	
Corrosion	3.2.4.5	4.4.4.4.6	
Fungistatic materials	3.2.4.6	4.4.4.4.7	
Waterproofness	3.2.4.7	4.4.4.4.8	b(1/50)
Design and Construction	3.3	4.4.4.5	
Materials, processes, and parts	3.3.1 thru 3.3.1.10	4.4.4.5.1	a(2.5)
Electro-magnetic Interference	3.3.2	4.4.4.5.2	
Identification marking	3.3.3 thru 3.3.3.2	4.4.4.5.3	a(6.5)
Workmanship	3.3.4	4.4.4.5.4	x
Interchangeability	3.3.5	4.4.4.5.5	x
Packaging	5.	4.4.3	a(6.5)

NOTES:

x - indicates an acceptance level of 100% for quality conformance lot-by-lot inspection.

a() - indicates a sampling AQL of () for quality conformance lot-by-lot inspection.

b() - indicates a control of () for quality conformance inspection.

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6.3.1 Control inspection at one in fifty (1/50). Except as specified in 6.3, control inspection as specified in TABLE III is suggested to be performed on one randomly selected assembly from each fifty assemblies consecutively produced except that not more than one test should be performed in a one month period nor less than one test in a three month period.

6.3.1.1 Control inspection failure of one in fifty (1/50). Failure of the assembly to meet these specified control inspections shall be considered cause for rejection of the production quantity represented. When cause for failure is determined, necessary corrections shall be made and proven by testing three randomly selected assemblies from that production quantity. One of three assemblies shall be subjected to all specified inspections. Control inspection verification for the two remaining assemblies shall be limited to the parameters directly related to the failure cause and the parameters affected by the corrective action taken.

6.3.2 Control inspection frequencies other than one in fifty (1/50). Control inspection is suggested to be performed on one assembly at the intervals listed in Table III except that there shall be not more than one test performed in a three month period nor less than one test in a six month period. The assembly shall be randomly selected from either the first ten assemblies produced for the interval defined, or the last ten assemblies produced from the preceding interval. This unit may be one of the assemblies inspected per 6.3.1.

6.3.2.2 Control inspection failure of other than one in fifty (1/50). Failure of a control assembly to meet the specified inspections shall be considered cause to reject subsequent assemblies until deficiencies that caused the failure are corrected and proven by testing three consecutively produced assemblies. Control inspection verification for the assemblies shall be limited to the parameters directly related to the failure cause and the parameters affected by the corrective action taken.

6.4 First article. When first article inspection is required, the Procuring Contracting Officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first lot of production items, a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.3. The Procuring Contracting Officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.5 Waivers and deviations. The Procuring Contracting Officer shall coordinate all requests for waiver or deviation to this specification with Benet Laboratories and the appropriate product assurance activity.

6.6 Subject term (key word) listing.

Electrical Motors

Motor Assembly

Motor, Tank Turret

6.7 Drawings. Drawings listed in Section 2 of this specification under the heading "U.S. Army Armament Research, Development and Engineering Center" (ARDEC) may also include drawings prepared by, and identified as Watervliet Arsenal, Ordnance Corps, Weapons Command, or U.S. Army Armament Research and Development Command, U.S. Army Armament, Munitions and Chemical Command, etc. Technical data originally prepared by these activities are now under the cognizance of ARDEC.

6.8 Changes from previous issue. None. This is a new specification.

Custodian:

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Preparing Activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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